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Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords:

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Krishna Mishra
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2026-27 (ODD)**	**Course Name**	**Lab: Generative AI**
Semester	**7**	**Course Code**	**23ADS1706**
Roll No	**41**	**Name of Student**	**Krishna Mishra**

Practical Number	**5**
Course Outcome	CO1: Explain the fundamentals of Artificial Intelligence, Neural Networks, Natural Language Processing, and Transfer Learning used in Generative AI applications. CO2: Apply Generative AI models such as GANs, VAEs, Transformers, BERT, GPT, and T5to solve real-world AI and NLP problems. CO3: Analyze and optimize prompts using prompt engineering techniques for improving the performance of Large Language Models in different application domains. CO4: Design and develop Generative AI solutions by integrating Foundation Models, Retrieval-Augmented Generation (RAG), tokenization, fine-tuning, and chunking techniques for domain-specific applications.
Aim	To build a simple English-to-Hindi machine translation model using an Encoder–Decoder (sequence-to-sequence) architecture with LSTM.
Problem Definition	Build a simple English-to-Hindi (or English-to-Marathi) translation model using an Encoder–Decoder architecture.
Theory (100 words)	Machine translation converts a sentence from a source language into a target language. The sequence-to-sequence model has two parts. The encoder is an LSTM that reads the English sentence word by word and compresses it into a context vector (its final hidden and cell states). The decoder is another LSTM that is initialized with this context vector and generates the Hindi sentence one word at a time. Special tokens <start> and <end> mark the beginning and end of the target sentence. During training, the correct previous word is given to the decoder (teacher forcing); during translation, the decoder uses its own previous output.

Procedure and Execution (100 Words)

Algorithm:

- Start.
- Store English–Hindi sentence pairs and add <start> and <end> tokens to the Hindi sentences.
- Build separate vocabularies for English and Hindi and convert the sentences into integer sequences.
- Build the encoder (Embedding + LSTM) and keep its final hidden and cell states.
- Build the decoder (Embedding + LSTM + Dense Softmax) initialized with the encoder states.
- Train the model using teacher forcing.
- Create separate encoder and decoder inference models.
- Translate a new sentence by generating Hindi words one by one until <end> is produced.
- Stop.

Code:

```
import numpy as np
import tensorflow as tf
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Input, Embedding, LSTM, Dense

tf.keras.utils.set_random_seed(42)

pairs = [
    ("i am happy", "मैं खुश हूँ"),
    ("how are you", "आप कैसे हैं"),
    ("what is your name", "आपका नाम क्या है"),
    ("my name is aditya", "मेरा नाम आदित्य है"),
    ("i love india", "मुझे भारत से प्यार है"),
    ("good morning", "सुप्रभात"),
    ("thank you", "धन्यवाद"),
    ("where are you going", "आप कहाँ जा रहे हैं"),
    ("i am a student", "मैं एक छात्र हूँ"),
    ("this is my book", "यह मेरी किताब है"),
    ("he is my friend", "वह मेरा दोस्त है"),
    ("she is a teacher", "वह एक शिक्षिका है"),
    ("i like tea", "मुझे चाय पसंद है"),
    ("the water is cold", "पानी ठंडा है"),
    ("open the door", "दरवाज़ा खोलो"),
    ("come here", "यहाँ आओ"),
    ("i am hungry", "मुझे भूख लगी है"),
    ("we are friends", "हम दोस्त हैं"),
    ("what is this", "यह क्या है"),
    ("i live in nagpur", "मैं नागपुर में रहता हूँ"),
    ("i study engineering", "मैं इंजीनियरिंग पढ़ता हूँ"),
    ("see you tomorrow", "कल मिलते हैं"),
    ("have a nice day", "आपका दिन शुभ हो"),
    ("i am learning ai", "मैं एआई सीख रहा हूँ"),
    ("the weather is nice", "मौसम अच्छा है")
]

src_texts = [src for src, tgt in pairs]
tgt_texts = ["<start> " + tgt + " <end>" for src, tgt in pairs]
```

```
def build_vocab(sentences):
    words = sorted(set(" ".join(sentences).split()))
    word_to_index = {word: i + 1 for i, word in
enumerate(words)}
    index_to_word = {i: word for word, i in
word_to_index.items()}
    return word_to_index, index_to_word


src_w2i, _ = build_vocab(src_texts)
tgt_w2i, tgt_i2w = build_vocab(tgt_texts)

src_vocab = len(src_w2i) + 1
tgt_vocab = len(tgt_w2i) + 1


def encode(sentences, word_to_index, max_len):
    data = np.zeros((len(sentences), max_len), dtype="int32")

    for row, sentence in enumerate(sentences):
        for col, word in enumerate(sentence.split()):
            data[row, col] = word_to_index[word]

    return data


max_src = max(len(s.split()) for s in src_texts)
max_tgt = max(len(s.split()) for s in tgt_texts)

enc_X = encode(src_texts, src_w2i, max_src)
dec_full = encode(tgt_texts, tgt_w2i, max_tgt)

dec_X = dec_full[:, :-1]
dec_y = dec_full[:, 1:]


# Encoder
enc_in = Input(shape=(None,))
enc_emb = Embedding(src_vocab, 64)(enc_in)

_, h, c = LSTM(128, return_state=True)(enc_emb)


# Decoder
dec_in = Input(shape=(None,))
dec_emb_layer = Embedding(tgt_vocab, 64)
dec_lstm = LSTM(128, return_sequences=True,
return_state=True)
dec_dense = Dense(tgt_vocab, activation="softmax")

dec_seq, _, _ = dec_lstm(
    dec_emb_layer(dec_in),
    initial_state=[h, c]
)

output = dec_dense(dec_seq)

model = Model([enc_in, dec_in], output)
```

```
model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)

history = model.fit(
    [enc_X, dec_X],
    dec_y,
    epochs=300,
    batch_size=8,
    verbose=0
)

print("Training Accuracy:", round(history.history["accuracy"][-1], 2))


# Inference models
enc_model = Model(enc_in, [h, c])

state_h = Input(shape=(128,))
state_c = Input(shape=(128,))

dec_seq2, h2, c2 = dec_lstm(
    dec_emb_layer(dec_in),
    initial_state=[state_h, state_c]
)

dec_model = Model(
    [dec_in, state_h, state_c],
    [dec_dense(dec_seq2), h2, c2]
)


# Translation function
def translate(sentence):
    tokens = [
        src_w2i[word]
        for word in sentence.lower().split()
        if word in src_w2i
    ]

    seq = np.zeros((1, max_src), dtype="int32")
    seq[0, :len(tokens)] = tokens

    h_s, c_s = enc_model.predict(seq, verbose=0)
    word_id = tgt_w2i["<start>"]
    result = []

    for _ in range(max_tgt):
        probs, h_s, c_s = dec_model.predict(
            [np.array([[word_id]]), h_s, c_s],
            verbose=0
        )

        word_id = int(np.argmax(probs[0, -1]))
        word = tgt_i2w.get(word_id, "")
```

<table>
<tr><td></td><td>

```
        if word == "<end>" or word == "":
            break

        result.append(word)

    return " ".join(result)


# Test translations
test_sentences = [
    "how are you",
    "i am happy",
    "thank you",
    "i live in nagpur",
    "the weather is nice"
]

for sentence in test_sentences:
  print(sentence, "->", translate(sentence))
```

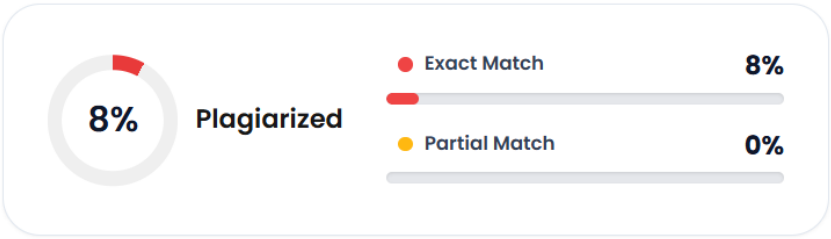
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<tr><td>Output Analysis</td><td>

- Training:
  The model is trained on 25 English–Hindi sentence pairs for 300 epochs and reaches a training accuracy of 1.0.
- Translations:
  The Hindi output for the sample English sentences is:
  how are you → □ □ □ □ □ □□ □ □
  i am happy → □ □ □□ □ □□ □ □
  thank you → □ □ □ □ □ □ □
  i live in nagpur → □ □ □□ □ □ □ □ □ □ □□ □ □ □□ □ □
  the weather is nice → □ □ □ □□ □ □ □ □□ □
- The model translates the trained sentences correctly. Because the dataset is very small, it cannot translate sentences with unseen words or structures; a larger parallel corpus is required for real translation.

</td></tr>
<tr><td>Link of student Github profile where lab assignment has been uploaded.

Plag Report (Similarity index < 12%)</td><td>

Github.com/krish-na45/GenAI-Practicals



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</table>

Conclusion	The experiment implemented an Encoder–Decoder model for English-to-Hindi translation. The encoder summarized the English sentence into a context vector and the decoder generated the Hindi sentence word by word. The practical explains sequence-to-sequence learning; accuracy on general text can be improved with a large dataset and attention or transformer models.
Date	/ /2026

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